Task 4: Optimal Externalization Threshold

The externalization strategy externalizes a trade if and only if the adversity model's predicted probability exceeds a cutoff θ:

$$\text{externalize} \iff p^{+/-} > \theta.$$

An externalized trade contributes exactly zero net PnL (the external hedge exactly offsets the internal position). For each client and horizon τ, the optimal threshold θ^* is selected by sweeping $\theta \in [0, 1]$ in steps of 0.01 on the validation set and choosing the value that maximises total validation PnL.

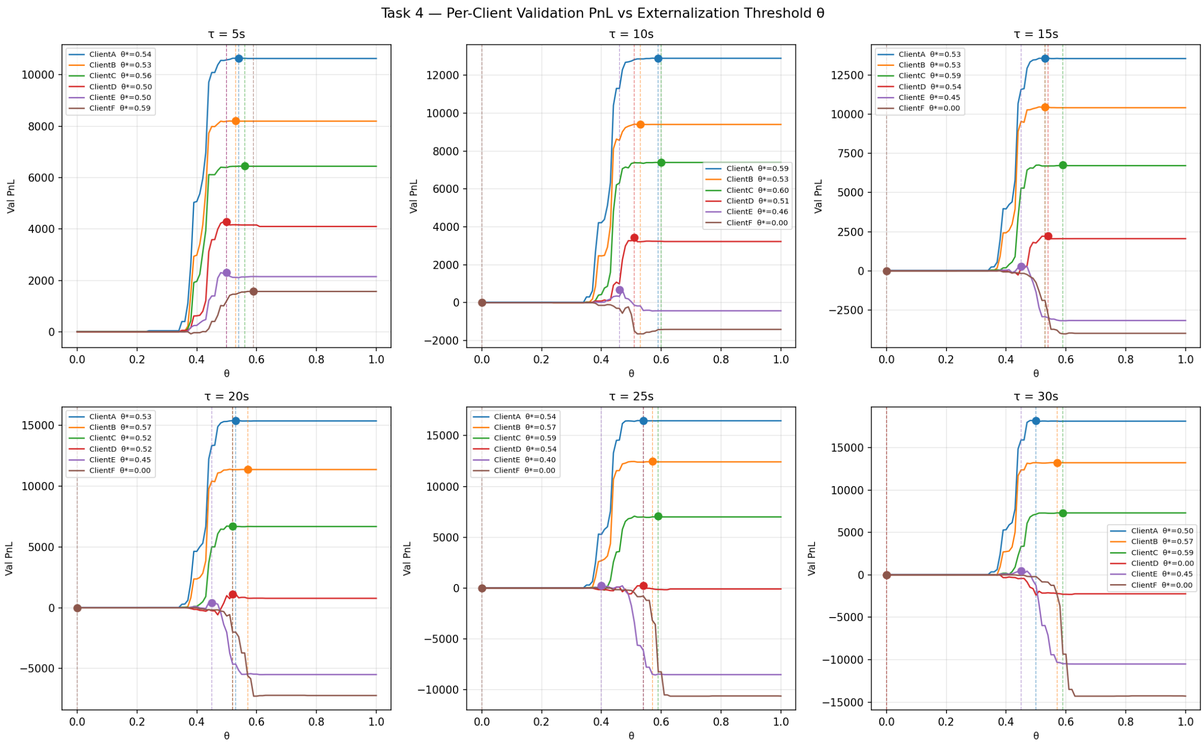

Figure 1: Per-client validation PnL as a function of the externalization threshold θ, for each horizon $\tau \in \{5, 10, 15, 20, 25, 30\}$. Filled circles mark the optimal θ^* for each client. The characteristic shape — flat near zero, a peak, then a decline — is consistent across horizons, but the peak location and height vary substantially by client.

Client	$\tau = 5$	$\tau = 10$	$\tau = 15$	$\tau = 20$	$\tau = 25$	$\tau = 30$
A	0.54	0.59	0.53	0.53	0.54	0.50
B	0.53	0.53	0.53	0.57	0.57	0.57
C	0.56	0.60	0.59	0.52	0.59	0.59
D	0.50	0.51	0.54	0.52	0.54	0.00
E	0.50	0.46	0.45	0.45	0.40	0.45
F	0.59	0.00	0.00	0.00	0.00	0.00

Table 1: Optimal threshold θ^* per client and horizon. A value of 0.00 denotes full externalization (all trades hedged away); this is optimal for Client F at $\tau \geq 10$ and Client D at $\tau = 30$, where internalising any flow yields a net loss.

Over- and Under-Externalization Trade-offs

Over-externalization ($\theta \to 0$): the LP externalizes nearly every trade, including profitable uninformed flow. Because externalized trades contribute exactly zero PnL, total revenue collapses toward zero — the LP effectively exits the market, foregoing all spread income from Clients A, B, and C. This is visible in Figure 1: for $\theta < 0.3$ all curves converge near zero.

Under-externalization ($\theta \to 1$): the LP internalizes almost everything, including toxic informed flow. For Clients E and F — whose adversity rises steeply with τ (Task 1) and whose aggregate PnL is negative at zero spread (Task 2) — this produces large losses. This is visible in Figure 1 as a sharp decline at high θ, most pronounced at longer horizons.

The optimal θ^* balances these two forces: high enough to retain profitable uninformed flow from Clients A–C, yet low enough to hedge the most toxic trades from Clients E and F.

Client-Specific Thresholds

The 36-cell optimisation reveals that θ^* varies meaningfully across clients, and a single global threshold would be sub-optimal:

- **Clients A, B, C (uninformed):** θ^* is consistently high (0.50–0.60) and stable across τ. The adversity model correctly identifies most of their flow as benign; a high threshold retains almost all trades.
- **Client D (borderline):** θ^* is moderate (0.50–0.54) at short horizons but drops to 0.00 at $\tau = 30$, meaning full externalization is optimal: the client turns loss-making at long horizons (Task 2) and no $\theta > 0$ improves on full hedging.
- **Client E (costly):** θ^* is lower (0.40–0.50) and falls with τ, reflecting the increasing adversity signal at longer horizons. Even so, some residual losses remain unavoidable.
- **Client F (most adverse):** $\theta^* = 0.59$ at $\tau = 5$ but drops to 0.00 for all $\tau \geq 10$. The validation PnL landscape is maximised by fully externalizing all medium- and long-horizon flow from this client.

The decrease in θ^* with τ for the informed clients (E, F) reflects the fact that adversity is higher and more predictable at longer horizons (Task 1), making earlier externalization worthwhile. Clients A–C exhibit no such trend.

Final Test-Set PnL

Using the per-client, per-τ optimal θ^* from validation, the strategy is applied to the held-out test set (days 34–42):

Client	$\tau = 5$	$\tau = 10$	$\tau = 15$	$\tau = 20$	$\tau = 25$	$\tau = 30$	**Row total**
ClientA	11,051	13,492	14,556	17,511	19,723	20,921	97,254
ClientB	9,428	10,577	11,829	12,197	13,937	15,057	73,025
ClientC	8,655	9,424	9,661	9,408	9,693	9,625	56,467
ClientD	4,822	3,555	2,053	87	−1,132	0	9,385
ClientE	3,392	153	−474	−565	−152	−426	1,927
ClientF	2,261	0	0	0	0	0	2,261
Col. total	39,609	37,202	37,625	38,637	42,069	45,177	**240,319**

Table 2: Test-set PnL ($) per client and horizon. Zeros for Client F ($\tau \geq 10$) and Client D ($\tau = 30$) reflect full externalization ($\theta^* = 0.00$). Negative cells for Clients D–E at longer horizons reflect residual adverse flow that the model correctly retains at intermediate thresholds but cannot fully eliminate.

The total test-set PnL across all clients and horizons is **$240,319**, compared to a baseline of $0 under full externalization. The bulk of the value (≈94%) comes from Clients A, B, and C — the uninformed clients whose flow the strategy correctly retains. Client F contributes only \$2,261 (all at $\tau = 5$), with zero contribution at longer horizons due to full externalization. Residual losses from Clients D, E, and F are small relative to gains from A–C, confirming the strategy successfully isolates and hedges toxic flow.